HK IMO Training 2023–2024
Problem Set 17
For 17 February 2024

Problem 65. In $\triangle ABC$, $AB < AC < BC$. Let Γ_1 be the circumcircle of $\triangle ABC$, let Γ_2 be the circle with centre B and radius AC, let Γ_3 be the circle with centre C and radius AB. Suppose D is an intersection point of Γ_2 and Γ_3, E is an intersection point of Γ_1 and Γ_2, F is an intersection point of Γ_1 and Γ_3, such that D, E, F lie on the opposite side of BC as A. Prove that the circumcentre of $\triangle DEF$ lie on Γ_1.

Problem 66. Find all positive integers m and n such that all digits of $(2m)^n + 1$ in the decimal representation are equal.

Problem 67. There are 110 participants in a tournament. Every pair of participants plays against each other exactly once, and there is no draw in each game. After all the games are played, it is found that for any 55 participants, there is one of them who has lost to at most 4 of the other 54 participants. Prove that there is a participant who has lost to at most 4 of the other 109 participants in the tournament.

Problem 68. Let $f : \mathbb{R} \to \mathbb{R}$ be a monotonic increasing function such that

$$f(y) - f(x) < y - x$$

for all $y > x$. Consider any sequence $a_1, a_2, \ldots$ of real numbers satisfying

$$a_{n+2} = f(a_{n+1}) - f(a_n)$$

for all $n \geqslant 1$. Prove that for any $\varepsilon > 0$, there exists a positive integer N such that $|a_n| < \varepsilon$ for all $n \geqslant N$.